Complete finite exclusions for integer equal-area rectangle partitions

Private recovery manuscript for L018

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This paper recovers the principal finite conclusions of García-Colín, Leemans, Müßig and Roldán [1], using the complete companion *Boundary constraints for equal-area rectangle dissections* and the whole finite programs and receipts specified below. It also records corrections to particular examples and implementation conventions. It does not assert that every literal source statement is true. The source's question about an integer perfect Mondrian partition of any size remains unanswered here.

Theorem 1 (Complete bounded conclusions)**.** *There is no axis-parallel partition of an integer-sided square of side at most* 1000*, or of an integer-sided rectangle whose two sides are at most* 400*, into at least two pairwise noncongruent integer-sided rectangles of equal area. Congruence includes a quarter-turn.*

Proof. By the companion's seven-piece theorem, any such partition has $r \geq 7$ pieces. Their common integer area is $A = WH/r$, so r is a divisor of WH. Each shape is an unordered pair $a \leq b$ with $ab = A$ and a fitting orientation. Interchanging the two axes if needed, take $W \geq H$; the exact fitting condition is $a \leq H$ and $b \leq W$. There must be at least r such shapes. The exact divisor and factor-pair enumeration in `enumerate_candidates.py` gives

range	containers with an arithmetic candidate	(W, H, r) cases
$W = H \leq 1000$	49	103
$H \leq W \leq 400$	6431	11126

All other parameter choices are excluded by these necessary conditions. The complete roster is stored explicitly, and its checker reconstructs it from every container and every divisor, rather than trusting the displayed counts.

Every listed case has a completed exhaustive-search exclusion. The companion proves that the finite skyline search follows any actual tiling and proves all prunings and symmetry reductions used here. The exact per-case route in `search_coverage.json` names the whole program, successful receipt, record index, and actual node count. It contains exactly the roster and no unresolved case. Earlier bounded runs marked `UNKNOWN` are retained as history, but no such record supplies an exclusion. This completed computation is an explicit finite proof premise, with ordinary compiler and machine-execution trust; it is not a kernel-checked proof or merely a sample of placements. □

The two largest square cases. The general exact search closes every square case except $(840, 840, 20)$ and $(840, 840, 21)$. For these, the two complete specialized programs add exactly the companion's proved parity reductions. In the 21-piece case, all 21 shapes must be used. Four have an odd side; the normalized even partners are $7, 5, 3, 1$. The alternating sums force their odd dimensions onto a common axis, their interval endpoints give six relative-height patterns, and their starting horizontal parities separate $7 + 1$ from $5 + 3$. In the 20-piece case, exactly

eight of the nine odd-sided shapes must be selected. Literal signed-subset arithmetic gives 44 configurations, or 26 after choosing the axis with at least four odd dimensions.

Both specialized programs retain the least-index bottom-left corner convention, using the axis-preserving reflections. They *drop* the independent bottom-left width-at-most-height convention, since the quarter-turn has already been used to normalize the odd dimensions. All corner comparisons are non-strict. The two completed runs respectively explore 77 756 396 and 59 729 842 states and return `INFEASIBLE`. Their observed times, about 41.8 and 31.9 seconds on this machine, are historical run measurements and are not complexity bounds. The distinct signed-sum/endpoint checker reconstructs the six and 44/26 lists and their reflection/rotation closure independently.

Whole finite dependency closure. The required executable sources are the complete divisor-roster generator, five complete C++ search versions, and the complete receipt-coverage assembler. The five versions preserve the exact bytes that produced the selected receipts: original lowest-column search, added side-subset pruning, most-constrained valley order, and the two specialized parity searches. The whole roster and all ten untrimmed receipt files are required mathematical data. The current assembly, metadata, binding and render driver is operational. The independent small-cell oracle, congruence-relaxed positive tests, source-example checker, and coloring and specialization checks are bounded validation; they do not replace the completed finite searches. Earlier exploratory CP-SAT models are not premises. No new Lean theorem is claimed, and no external source census is assumed.

Source examples and corrected perimeter convention. The first square with enough shapes is 84 with $r = 7$ and common area 1008. Its unordered shape list is

$$(12, 84), (14, 72), (16, 63), (18, 56), (21, 48), (24, 42), (28, 36).$$

Several products in the printed list on physical page 4 differ from 1008; the list above is the exact factor-pair replacement. The bottom-side example $12 + 72 = 84$ remains an example of partial boundary coverage.

The statement that a possible perimeter first occurs at 360 needs an extra no-spanning convention. An actual five-piece equal-area, noncongruent perimeter in the 84-square is given by the following (x, y, w, h) coordinates:

$$(72, 0, 12, 84),\ (0, 0, 16, 63),\ (16, 0, 56, 18),$$
$$(0, 63, 48, 21),\ (48, 42, 24, 42).$$

These interiors are disjoint and cover all four sides; the first piece spans the full height and occupies two corners. This is not a perfect partition. Removing its spanning strip would leave a six-piece perfect partition of a rectangle if a seven-piece completion existed, contrary to the companion's seven-piece theorem. A correct general perimeter test allows opposite sides to share a spanning piece and repeated corner identities, or explicitly pays for a valid recursive exclusion. One cannot silently require four distinct corner pieces in a search for arbitrary partitions.

For the source's 360-square example, the seven perimeter pieces have coordinates

$$(0, 0, 240, 45),\ (240, 0, 120, 90),\ (0, 180, 60, 180),$$
$$(60, 252, 100, 108),\ (160, 306, 200, 54),$$
$$(0, 45, 80, 135),\ (310, 90, 50, 216).$$

Each area is 10800. The five unused shapes are $(36, 300), (40, 270), (48, 225), (72, 150), (75, 144)$. The printed assertion that their dimensions cannot sum to 216 is false: $72 + 144 = 216$ uses two distinct listed shapes. This arithmetic possibility is not a tiling. The other printed arithmetic

equality $40 + 48 + 72 = 160$ is correct, but a 270-high piece above the bottom gap at height 45 reaches 315 and intersects the top perimeter, whose lower edges are all at most 306.

The hole bound correctly excludes this perimeter regardless of that mistaken subset sum. Complete left and right boundary coverings occupy vertical strips of widths equal to their respective minimum widths. The corresponding top and bottom strips are also occupied. Every piece not on the perimeter must therefore fit in the bounding box

$$(360 - 60 - 50) \times (360 - 45 - 54) = 250 \times 261.$$

Neither orientation of the required 36×300 piece fits. This is a containing box, not an assertion that the interior itself is rectangular. All displayed coordinates, areas, disjointness, side coverings and arithmetic witnesses are checked in `source_examples.py`.

Scope of method reports and earlier results. The stronger complete exclusions here imply the source's numerical lower ranges 360 and 420 without equating the independent search with the historical sequence of side, perimeter, gap and hole implementations. Its eleven-side filtered list and runtime heatmap are retained as reports about that implementation, not borrowed premises of this proof or newly reproduced method counts. The source's current public Julia revision is pinned separately. The example $W = 60, H = 14, r = 7, A = 120$ has seven fitting unordered shapes but eight fitting orientations, so half the filtered orientation count is not the shape count and reversal of that filtered list is not generally the quarter-turn map. This audits that exact current revision; no claim about the historical execution revision or a counterexample to the complete bounded conclusions follows.

The source's prior real dissections, earlier minimum-defect computations through 65, constructions through 96, density estimates and stochastic-process links remain expressly attributed background, not unproved premises of either bounded theorem here. The elementary defect bounds follow by two full-height strips: widths $(n-1)/2, (n+1)/2$ give defect n for odd $n \geq 3$; widths $n/2 - 1, n/2 + 1$ give defect $2n$ for even $n \geq 4$. The nontrivial-partition convention excludes $n = 1, 2$ from these constructions. The general integer-existence question and the stated logarithmic defect-bound conjecture remain unanswered, with no new question, formalization or compression credit.

References

[1] N. García-Colín, D. Leemans, M. Müßig and É. Roldán, *There is no perfect Mondrian partition for squares of side lengths less than 1001*, arXiv:2311.02385v1, eleven-page fixed source edition (2023).